

Liquid-Liquid Extraction

PRE-LAB PLAN

Unit Operations Lab 4

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1. Experimental Objective

The objectives of this experiment are to develop plots of acetic acid concentration as a function of stage number and agitator speed, plotting acetic acid concentration in the water versus the concentration in oil, calculating the number of theoretical stages using McCabe-Thiele analysis, obtaining the Murphee efficiencies for the stages, and determining the overall efficiency of the set up. This is accomplished by experimenting with various agitator speeds; different data points are recorded for calculation purposes.

2. Experimental

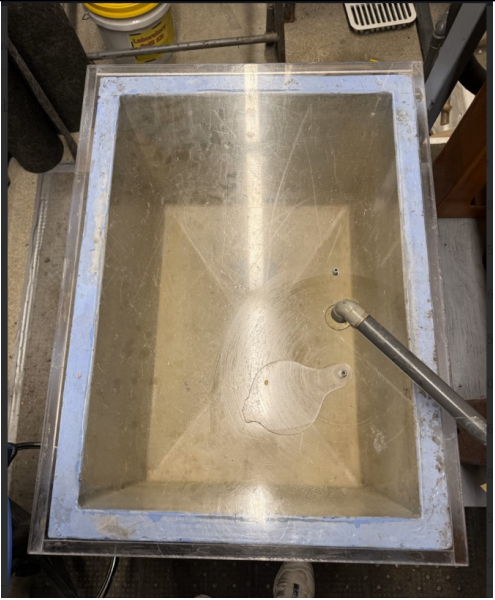
This experiment involves analyzing the concentration of acetic acid within water and mineral oil through three different agitator speeds. For this experiment, the necessary data include agitator speed, water flowrate, oil flowrate, amount of NaOH needed for titration, oil sample volume, and water sample volume. The experiment also uses a variety of equipment and materials, which are summarized below

2.1 Apparatus

<i>Number</i>	<i>Name</i>	<i>Main vs Auxiliary</i>	<i>Description</i>	<i>Image</i>
1	[1] Three-Stage Denver mixer-settler liquid-liquid extraction unit [2] Individual Stage in Unit	Main	Unit where experimental procedure takes place	

2	[1] Agitator motors [2] Agitator knob	Main	To control mixing/rate of mixing in unit	
3	[1] Feed tank [2] Product tank	Main	[1] Houses the initial 25 gal. of oil for the experiment [2] Where the separated mix. is removed from the unit	

1

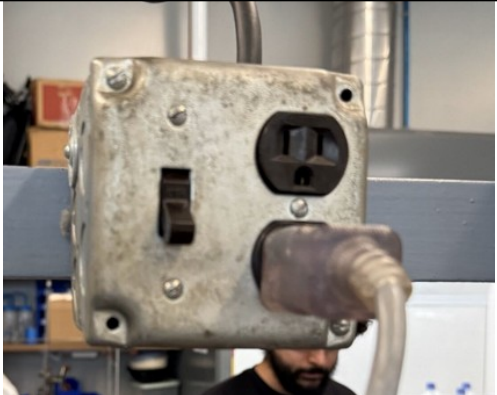
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4	[1] Water rotameter [2] Oil rotameter	Auxiliary	[1] To measure and read flow of water [2] To measure and read flow of feed	
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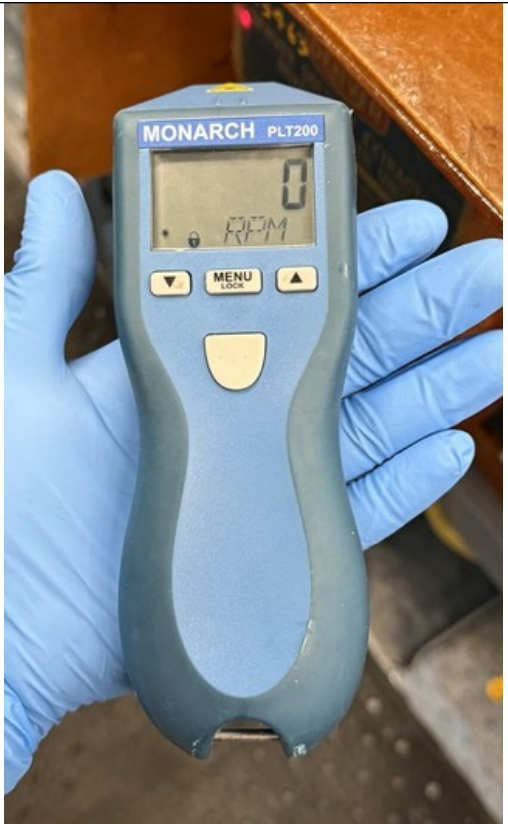
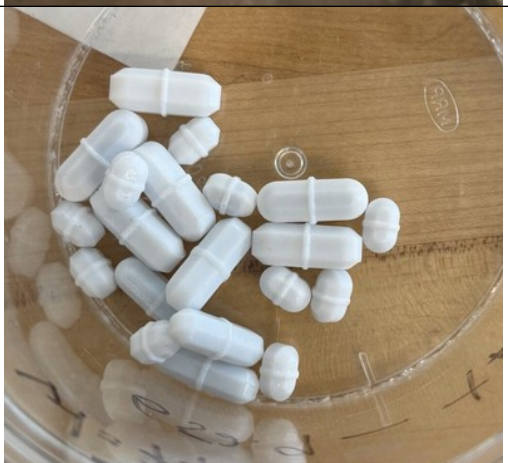


5	Pump	Main	To assist with flow of oil	 The image shows a black Dayton pump motor. The label on the motor provides the following specifications: - Model: P20183409 - Frame: 1 - HP: 1/2 - Volts: 230V - Phases: 3 - Frequency: 60 Hz - Speed: 1725 RPM - Service Factor: 1.15 - Weight: 15 lbs - Dimensions: 14" x 14" x 14" Below the specifications, there is a diagram of the motor's terminal box with the following labels: - Line Voltage - Line Connections - Grounding - Neutral - Ground A blue sticker from the University of Illinois is attached to the motor, with the text "UNIVERSITY OF ILLINOIS STATE OF ILLINOIS" and the number "P20183409".
6	Circuit Breaker for Agitators	Main	To help control power to Agitators	 The image shows a grey safety switch. The label on the switch reads: - SAFETY SWITCH - Model: P20183409 - Frame: 1 - HP: 1/2 - Volts: 230V - Phases: 3 - Frequency: 60 Hz - Speed: 1725 RPM - Service Factor: 1.15 - Weight: 15 lbs - Dimensions: 14" x 14" x 14"

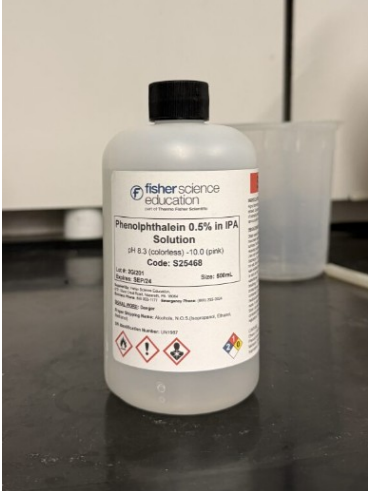
7	Pump Power switch	Main	Control power to pump	
8	Titration Setup	Main	To carry out titration when checking pH levels of separated mix.	

2.2. Materials and Supplies

Number	Name	Material /Supply	Description	Image
1	Water	Material	Used to make mix. for experiment	

2	Tachometer	Supply	Used to read RPM of agitators used in experiment	
3	Magnetic Stir Bars	Supply	Used in titration experiment to help with mixing indicator, acid and liquid being tested	

4	Covers for Unit Stages	Material	To cover each unit stage when carrying out procedure (three in total)	
5	Safety Goggles	Supply	Prevent hazardous chemicals from contact with eyes	

6	Gloves	Supply	Prevent hazardous chemicals from contact with skin	
7	Phenolphthalein	Material	Utilized in titration to identify amount of citric acid in sample	

2.3. Experimental Plan

Experimental Variables:

Independent Variables: Agitator Speed, Will be running the experiment at three different agitator speeds between 350RPM – 1000 RPM

Dependent Variables: Acetic acid concentration: Acetic acid concentration in water, and Acetic acid concentration in oil. Residence Time, flow rates, stage to stage compositions.

Controlled Variables: Solvent feeds, Flow rates, Solvent feed ratio: 2 to 1 (water to oil). ~1wt% acetic acid in oil

Setup:

1. Ensure that the feed tank contains ~25 gallons of mineral oil. If needed, have the TA pump the oil back into the feed tank.
2. Calculate the amount of acetic acid needed to add into the mineral oil.
3. Add the acetic acid into the oil and mix it using the centrifugal pump in a closed loop.
4. Check the initial acid in oil concentration by titration using NaOH using phenolphthalein.

Experiment:

1. Ensure all the drain valves underneath all the stages are closed.
2. Start the water flow to the extractor first and establish steady liquid levels in all stages.
3. Once it has reached steady liquid levels set the flow rate to 0.7 GPM
4. Turn on the agitators and set all the agitators to the first RPM setpoint using the tachometer.
5. Once the water circuit is stable, begin feeding the oil mixture by opening the feed valve and closing the return valve.
 - a. Also adjust the flowrate to 0.35 GPM for the feed
6. Keep flow rates the same throughout the entire experiment.
 - a. The flowrate on the feed will change slightly throughout the experiment so have someone nearby to adjust it back to 0.35 GPM
7. Let the unit run until the oil reaches the product tank. Record the time as this will be the residence time used for all three experiments.
8. Once steady state is reached (this will be about one residence time) take samples from both phases across the three stages.
9. Cover the samples in parafilm to prevent any acetic acid losses. Then titrate each of the samples using NaOH using phenolphthalein. While also using constant stirring during the titration.
10. Record the RPM, flow rates, residence time, and titration for every sample
11. Repeat steps 1-10 for the other two more agitator speeds. Keeping flow rates constant and drain valves closed.

Data Analysis:

- Compute acetic acid wt% in each of the samples.
- Plot concentrations vs. Stage numbers for each agitator speed.
- Plot concentrations vs. Different agitator speeds.
- Calculate Murphree efficiency across the three stages.
- Calculate overall efficiencies using actual stages and the theoretical stages from McCabe-Thiele.

- Compare how changing the RPM affects the concentration across all three stages.

Safety Analysis:

Hazard	Yes/No	If Yes, Description
Potential electrical hazards?	Yes	120V and 240V are supplied to the unit and could possibly come into contact with solutions present in the experiment posing risk
Potential mechanical hazards?	Yes	The rotating shafts of each of the 3 impellers along with the oil pump pose a risk to lose articles such as clothing, jewelry or hair.
Potential pressure hazards?	No	
Potential temperature hazards?	No	
Hazardous/toxic chemicals involved	Yes	Sodium hydroxide is a strong base, Glacial acetic acid is a concentrated acid, Phenolphthalein is a suspected human carcinogen
Gloves required?	Yes	
MSDS reviewed by all team members	Yes	Water (7732-18-5), Mineral Oil (8042-47-5), Glacial acetic acid (64-19-7), Sodium hydroxide solution (1310-73-2), Phenolphthalein (77-09-8)
Process parameters		
Temperature (°C)	21	Experiment is run at room temperature
Pressure (psia)	20	Present at the outlet of the pump
Safety controls	None	

I have read relevant background material for the Unit Operations Laboratory entitled: “Liquid-Liquid Extraction” and understand the hazards associated with conducting this experiment. I have planned out my experimental work in accordance to standards and acceptable safety practices and will conduct all of my experimental work in a careful and safe manner. I will also be aware of my surroundings, my group members, and other lab students, and will look out for their safety as well.

Alexis Guillen

Signature of team member

Khoa Nguyen

Signature of team member

3/3/26

Date

3/3/26

Date

Rishabh S. Patel	3/3/26
Signature of team member	Date
Kyle Hankosky	3/3/26
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